GLS University
FCAIT
iMSCIT SEM VI
221601605 Practicals on Machine Learning
Practical Assignment Unit3

1.

Dataset:

Age	Income	Credit Score	Loan Approved
25	3	650	No
30	5	700	Yes
45	8	720	Yes
35	4	680	No
50	10	750	Yes
28	3	640	No

Write a Machine Learning code to perform following:

- Convert Loan Approved Categorical data to Numeric Data
- Consider Age, Income and Credit Score as an input field and Loan Approved as an output
- Train a model using decision tree method.
- Test a model is Age is 30, Income is 8 Lacs and Credit Score is 700, Whether Loan approved or not.
- Visualize the data

2. Dataset:

Age	Blood Sugar (mg/dL)	BMI	Diabetes
22	85	21	No
35	140	26	Yes
50	160	30	Yes

28	95	23	No
45	155	29	Yes
32	100	24	No

Write a Machine Learning code to perform following:

- Create a csv file for above data
- Build a decision tree.
- Split the dataset into training and testing sets.
- Evaluate the model using **accuracy**.

3. Dataset:

Temperature	**Humidity**	**Wind Speed**	**Activity**
Hot	High	Weak	No
Hot	High	Strong	No
Mild	Normal	Weak	Yes
Cool	Normal	Strong	Yes
Mild	High	Weak	Yes
Cool	High	Strong	No

Write a Machine Learning code to perform following:

- Encode categorical variables appropriately.
- Train a decision tree classifier.
- Visualize the decision tree.
- Predict the activity for:
- Temperature = Mild, Humidity = Normal, Wind Speed = Strong

4. Write a Machine Learning program to implement Simple Linear Regression for predicting house prices based on size.

Size (sq ft)	Price ($)
500	50000
800	80000
1200	120000
1500	150000
1800	180000

5. Write a Machine Learning program using Multiple Linear Regression to predict house prices based on multiple factors like size, number of bedrooms, and age of the house.

Size (sq ft)	Bedrooms	Age (years)	Price ($)
1500	3	10	300000
1800	4	5	400000
2400	4	20	500000
3000	5	15	600000
3500	5	8	700000

6. Write a Machine Learning program using Multiple Linear Regression to predict car prices based on horsepower, mileage (MPG), and car age.

Horsepower	Mileage (MPG)	Age (years)	Price ($)
150	30	5	20000
180	25	3	25000
200	20	7	22000
250	18	2	30000
300	15	1	40000